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Experiment 1: Analysis of sound wave and Frequency Spectrum

Aim.

To record a temple bell sound, conch sound, or classical music note and analyze its time domain waveform and frequency spectrum to identify the fundamental frequency and harmonics.

Theory

Sound is produced due to vibrations and travels in the form of waves. A sound signal can be analyzed in two ways:

1. Time Domain: Shows how the amplitude of sound changes with time.
2. Frequency Domain: Shows the different frequencies present in the sound.

The lowest major frequency present in a periodic sound is called fundamental frequency.

Procedure

- Record a temple bell sound or a classical song note using a mobile phone.
- Import the recorded audio into a sound analysis software or Python program.
- Apply frequency analysis such as FFT to obtain the frequency spectrum.
- Identify the strongest lowest frequency as the fundamental frequency.
- Observe the higher-frequency peaks representing harmonics.

2. Compare the Observed freq. relations with
with theoretical music ratios
 $\frac{1}{2}, \frac{3}{4}, \frac{5}{6}, \frac{7}{8}, \frac{9}{10}, \frac{11}{12}, \frac{13}{14}, \frac{15}{16}, \frac{17}{18}, \frac{19}{20}, \frac{21}{22}, \frac{23}{24}, \frac{25}{26}, \frac{27}{28}, \frac{29}{30}, \frac{31}{32}, \frac{33}{34}, \frac{35}{36}, \frac{37}{38}, \frac{39}{40}, \frac{41}{42}, \frac{43}{44}, \frac{45}{46}, \frac{47}{48}, \frac{49}{50}, \frac{51}{52}, \frac{53}{54}, \frac{55}{56}, \frac{57}{58}, \frac{59}{60}, \frac{61}{62}, \frac{63}{64}, \frac{65}{66}, \frac{67}{68}, \frac{69}{70}, \frac{71}{72}, \frac{73}{74}, \frac{75}{76}, \frac{77}{78}, \frac{79}{80}, \frac{81}{82}, \frac{83}{84}, \frac{85}{86}, \frac{87}{88}, \frac{89}{90}, \frac{91}{92}, \frac{93}{94}, \frac{95}{96}, \frac{97}{98}, \frac{99}{100}$

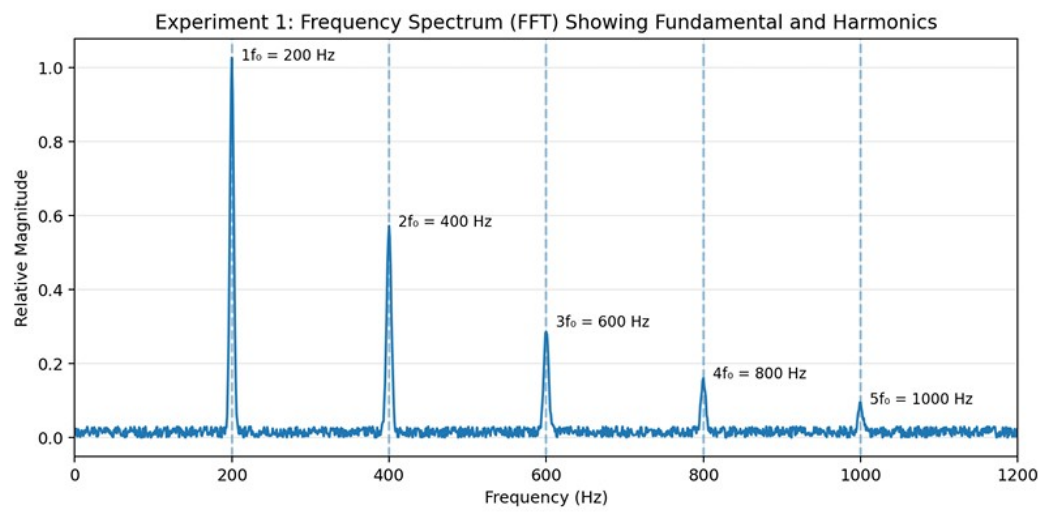
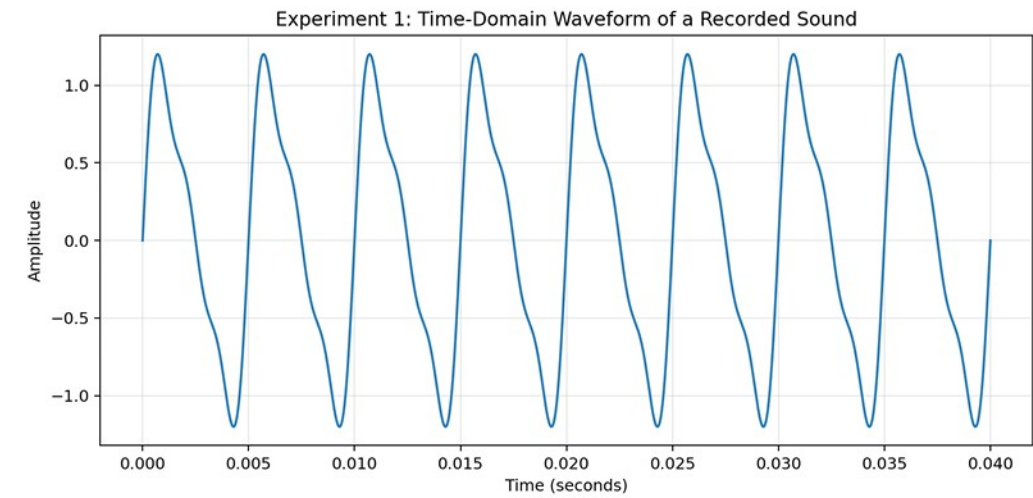
Observation

The waveform showed variations in amplitude with time. The frequency spectrum showed a dominant fundamental frequency component.

Result

The recorded sound was successfully analyzed in both time and frequency domains. The fundamental frequency and harmonic components demonstrated that complex natural sounds consist of multiple frequency components. This experiment also shows the connection between sound analysis, Indian musical knowledge and modern digital signal processing.

Experiment No. 1 – Analysis of Sound Wave and Frequency Spectrum



Component	Approx. frequency	Relationship
Fundamental (f_0)	200 Hz	f_0
2nd harmonic	400 Hz	$2f_0$
3rd harmonic	600 Hz	$3f_0$
4th harmonic	800 Hz	$4f_0$
5th harmonic	1000 Hz	$5f_0$